Brendon Peters

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Education

Worcester Polytechnic Institute — Worcester, MA

B.S. Computer Science, Minor in Data Science, GPA: 3.8 — *May, 2027*

M.S. Artificial Intelligence, GPA: 4.0 — *December 2027*

Involvement: IEEE: Vice President, ACM, Data Science Club, High Power Rocketry Club, Tennis Club

Relevant Coursework: Deep Learning, Computer Vision, Artificial Intelligence, Robotics, Machine Learning, Software Engineering, Operating Systems, Embedded Systems, Data Structures and Algorithms, Database Systems

Experience

Software Researcher (MQP) — August 2026 – Present

Microsoft — *Worcester, MA*

- Conduct extensive background research into graph-based AI reasoning, evaluating Neo4j to map indirect, multi-step dependencies within Microsoft Azure environments.
- Model complex interactions between system configurations, hardware resources, and operational states to support enterprise administrative decision-making.
- Develop custom benchmarks to evaluate the speed and accuracy of graph data navigation strategies compared to traditional relational data models.

Software Developer — August 2024 - Present

Worcester Polytechnic Institute Mechanical Engineering Dept. — *Worcester, MA*

- Collaborated in a team of 3 to enhance the BoGL web application, improving bond diagram generation for mechanical, electrical, and hydraulic systems.
- Designed, implemented, and maintained new features using C#, TypeScript, HTML5, and CSS to extend the application's functionality.

Software Engineering Intern — June 2025 - August 2026

Advanced Acoustic Concepts — *Great River, NY*

- Engineered embedded control systems for underwater unmanned vehicle platforms, emphasizing performance and user experience.
- Developed autonomous control logic integrating visual and sonar sensor inputs to enhance tracking accuracy and operational reliability in underwater environments.
- Reduced operational bandwidth of remote systems and communication latency by 15% and 23%, respectively.

Projects

Web Accessibility Research (IQP) | *WordPress, Research, User Testing* — January 2026 – May 2026

- Partnered with Videnscenter om Handicap and Microsoft Denmark to engineer a self-hosted WordPress educational platform featuring AI-powered tools and strict EU/WCAG/POUR frontend compliance.
- Directed iterative usability testing with target users via a think-aloud protocol, analyzing quantitative performance metrics and qualitative feedback to inform UI/UX enhancements.

Brigham & Women's Hospital Website | *Typescript, React.js, PostgreSQL, Express* — March 2025 - May 2025

- Led full-stack development of a cloud-based hospital web app supporting indoor pathfinding, map editing, and service request workflows using the PERN stack, TypeScript, TailwindCSS, Shadcn/UI, and Prisma ORM.
- Built key features including a graph generator from floor plan images and a voice-activated interface; deployed via Docker and AWS, and led a 10-person Agile team through technical planning and delivery.

Web & Data Tool Development for Poverty Metrics | *Python, Flask, Django* — March 2025 - May 2025

- Collaborated with Fundación Paraguaya on the Poverty Stoplight project to develop a web platform comparing Multidimensional Poverty Index (MPI) and Poverty Stoplight (PSL) indicators.
- Engineered a data platform to compare MPI and PSL indicators, structuring PSL data for LLM fine-tuning to improve reasoning on multidimensional poverty frameworks.

Revalatio Rocket Payload | *C++, Github, Arduino* — August 2023 - June 2024

- Designed C++ software for an image-processing, self-landing rocket payload achieving a 15,000-foot apogee, securing multiple awards at the Intercollegiate Rocketry Engineering Competition against 150+ teams.

Skills

Programming Languages: Python, C/C++, MATLAB, Java, SQL, C#, R
AI/ML Frameworks: PyTorch, NumPy, Pandas, Scikit-learn
Tools & Technologies: ROS 2, Linux, Git, Subversion (SVN), Docker, Azure, TCP/UDP, PostgreSQL